

Test Certificate

Product Code:	TVBQ1FAA004YW
Description:	4f, Riser FRP/ARAMID/LSZH
Cable #:	13364
Date Cert created:	10 th Feb 2022
Design Number:	47-20
Product Family:	TVBQ**AA04xx

TEST PARAMETER / METHOD	AFL ACCEPTANCE CRITERIA	ACHIEVED	RESULT
Tensile Strength IEC 60794-1-21 (E1)	1kN 30min, < 0.6% fibre strain, No att. Increase after test, < 0.1dB during test, @ 1550nm	1kN <0.6% fs $\Delta\alpha \leq 0.10\text{dB}$ No att. Increase after test	Pass
Bend under tension IEC 60794-1-21 (E18A)	$r_b = 20 \times \text{OD}$ 360°, 1 turn, 1 min No fibre break, no damage to sheath or core structure and No attn. change throughout test	1kN $r_b = 20 \times \text{OD}_{\text{measured}}$ No breaks or damage and no attenuation change	Pass
Crush (short term) IEC 60794-1-21 (E3)	1kN/100mm Duration: 10min 3 position No damage to sheath or core structure and No att. Increase after test	>1kN/100mm No damage and no attenuation increase	Pass
Crush (long term) IEC 60794-1-21 (E3)	0.5kN/100mm Duration: 120min 3 positions No damage to sheath or core structure and No att. Increase during or after test	>0.5kN/100mm No damage and no attenuation increase	Pass
Impact IEC 60794-1-21 (E4)	Weight: 0.5kg Height: 1.0m, 300mm rad anvil, 3 impacts, After 5 mins No fibre break, no damage to sheath or core structure and No attn. change throughout test	No breaks or damage and no attenuation change	Pass
Bend IEC 60794-1-21 (E11)	$r_b = 10 \times \text{OD}$ 360° loop, 1 turn, 1 cycle No att. Change throughout test	$r_b = 10 \times \text{OD}$ No attn. Change	Pass
Torsion IEC 60794-1-21-(E7)	Sample length: 1 m Rotation: a) 180° clockwise, b) return to starting position, c) 180° anticlockwise, d) return to starting position. Four movements = one cycle. Complete each cycle within 1 min max. for total 10 cycles (a to d) During the final tenth cycle at a), c) and after completion (no rotation) check transmitting fibres. No fibre breaks, no damage to the sheath or to the core structure and no attenuation change throughout test	No breaks or damage and no attenuation change	Pass
Temperature Cycle IEC 60794-1-22 (F1)	Temperature Range: -10°C to +70°C $\Delta\alpha_{\text{max}} \leq 0.15\text{dB}$ $\Delta\alpha_{\text{avg}} < 0.05\text{dB}$ @ 1550nm	Temperature Range: -10°C to +70°C $\Delta\alpha_{\text{max}} \leq 0.15\text{dB}$ $\Delta\alpha_{\text{avg}} < 0.05\text{dB}$ @ 1550nm	Pass
Vertical Flame Test IEC 60332-1	Cable Must self-extinguish, No falling particles to ignite tissue paper, Damage/Carbonization distance from lower edge of top support: must not extend upwards $\geq 50\text{mm}$ and must not extend downwards $\leq 540\text{mm}$	Cable Must self-extinguish No falling particles to ignite tissue paper Damage/Carbonization must only reach 50mm max. under upper fixing	Pass
Water Penetration IEC 60794-1-22 (F5B)	Sample length: 3m Water Height: 1m No water after 24 hours	No Water detected	Pass
Tested By:	Attila N & Karlo L	Verified By:	Shashini Subasinghe
Approved By:	Jim Boukouvalas		